
Redyx — Comment Service Test Cases

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Comment Service — Test Case Specification

Project Name	Redyx
Module Name	Comment
File	internal/comment/server_test.go
Total Tests	18
Package	comment (same-package test)
Created By	idityaGE
Date of Creation	2026-05-10

Test Infrastructure

Component	Implementation
Logger	<code>zap.NewNop()</code>
Store	<code>nil</code> — tests short-circuit before store access
External Clients	<code>nil</code> — not required for validation tests
Auth Context	<code>auth.WithClaims(context.Background(), &auth.Claims{...})</code>
Helper convention	<code>t.Helper()</code> in <code>testCommentServer</code> , <code>authedCtx</code>
Nil recovery	<code>defer recover()</code> for tests that validate pass-through to nil store

1. mapSortOrder Tests

Test Case ID	Test Name	Pre-Condition	Test Steps	Test Data	Expected Result	Status
TC_CMNT_001	TestMapSortOrder_AllCases		Call <code>mapSortOrder</code> for each proto enum value	BEST, TOP, NEW, CONTROVER-SIAL, UNSPECIFIED (5 sub-cases)	BEST→SortBest, TOP→SortTop, NEW→SortNew, CONTROVER-SIAL→SortControversial, UNSPECI-FIED→SortBest (default fallback)	PASS

2. CreateComment Validation Tests

Test Case ID	Test Name	Pre-Condition	Test Steps	Test Data	Expected Result	Status
TC_CMNT_002	TestCreateCommentUnauthenticated	Comment server initialized	Call CreateComment with unauthenticated context	postId = "post-1", body = "A valid comment body"	Returns an error (missing auth claims)	PASS
TC_CMNT_003	TestCreateCommentEmptyPostID	Authenticated context	Call CreateComment with empty post_id	postId = "", body = "A valid comment body"	Returns an error	PASS
TC_CMNT_004	TestCreateCommentEmptyBody	Authenticated context	Call CreateComment with empty body	postId = "post-1", body = ""	Returns an error	PASS
TC_CMNT_005	TestCreateCommentBodyTooLong	Authenticated context	Call CreateComment with 10 001-character body	body = 10 001×"a"	Returns an error (max 10 000 chars)	PASS

3. GetComment Validation Tests

Test Case ID	Test Name	Pre-Condition	Test Steps	Test Data	Expected Result	Status
TC_CMNT_006	TestGetCommentEmptyCommentID	Comment server initialized	Call GetComment with empty comment_id	commentId = ""	Returns an error	PASS

4. UpdateComment Validation Tests

Test Case ID	Test Name	Pre-Condition	Test Steps	Test Data	Expected Result	Status
TC_CMNT_007	TestUpdateCommentUnauthenticated	Comment server initialized	Call UpdateComment with unauthenticated context	commentId = "comment-1", body = "Updated body"	Returns an error	PASS

Test Case ID	Test Name	Pre-Condition	Test Steps	Test Data	Expected Result	Status
TC_CMNT_008	TestUpdateCommentEmptyCommentID	Authenticated context	Call UpdateComment with empty comment_id	commentId = "", body = "Updated body"	Returns an error	PASS
TC_CMNT_009	TestUpdateCommentEmptyBody	Authenticated context	Call UpdateComment with empty body	commentId = "comment-1", body = ""	Returns an error	PASS
TC_CMNT_010	TestUpdateCommentBodyTooLong	Authenticated context	Call UpdateComment with 10 001-character body	body = 10 001×"x"	Returns an error (max 10 000 chars)	PASS

5. DeleteComment Validation Tests

Test Case ID	Test Name	Pre-Condition	Test Steps	Test Data	Expected Result	Status
TC_CMNT_011	TestDeleteCommentUnauthenticated	Comment server initialized	Call DeleteComment with unauthenticated context	commentId = "comment-1"	Returns an error	PASS
TC_CMNT_012	TestDeleteCommentEmptyCommentID	Authenticated context	Call DeleteComment with empty comment_id	commentId = ""	Returns an error	PASS

6. ListComments Validation Tests

Test Case ID	Test Name	Pre-Condition	Test Steps	Test Data	Expected Result	Status
TC_CMNT_013	TestListCommentsEmptyPostID	Comment server initialized	Call ListComments with empty post_id	postId = ""	Returns an error	PASS
TC_CMNT_014	TestListCommentsValidRequest_FailsAtStore	Nil store	Call ListComments with valid post_id and limit > 50	postId = "some-post-id", limit = 200	Validation passes (limit clamped to 50); panics at nil store (recovered) or returns store error	PASS

7. ListReplies Validation Tests

Test Case ID	Test Name	Pre-Condition	Test Steps	Test Data	Expected Result	Status
TC_CMNT_015	TestListRepliesCommentEmptyCommentID	server initialized	Call ListReplies with empty comment_id	commentId = ""	Returns an error	PASS

8. ListCommentsByAuthor Validation Tests

Test Case ID	Test Name	Pre-Condition	Test Steps	Test Data	Expected Result	Status
TC_CMNT_016	TestListCommentsByAuthorEmptyUsername	server initialized	Call ListCommentsByAuthor with empty username	username = ""	Returns an error	PASS

9. commentToProto Conversion Tests

Test Case ID	Test Name	Pre-Condition	Test Steps	Test Data	Expected Result	Status
TC_CMNT_017	TestCommentToProtoValidUUIDsNonZeroParentID_SetInProto	Valid UUIDs for all IDs	Construct Comment with non-zero ParentID; call commentToProto	parentId = "cccc...", commentId = "aaaa...", body = "Hello world", voteScore = 5, authorUsername = "alice"	proto.ParentId is non-empty; CommentId, Body, VoteScore, AuthorUsername all match source	PASS
TC_CMNT_018	TestCommentToProtoZeroParentID_NotSetInProto	Valid comment/post/author UUIDs; ParentID = zero UUID	Construct Comment without ParentID; call commentToProto	parentId = zero UUID (top-level comment)	proto.ParentId is empty string (top-level has no parent)	PASS

Patterns & Conventions

Pattern	Implementation
Package scope	<code>package comment</code> (white-box / same-package)
Logger	<code>zap.NewNop()</code> (silent)
Nil store	<code>NewServer(nil, nil, nil, zap.NewNop())</code> — validation-only
Panic recovery	<code>defer recover()</code> guards in tests that let execution reach nil store
Auth context	<code>auth.WithClaims(context.Background(), &auth.Claims{UserID, Username})</code>
Helper tagging	<code>t.Helper()</code> in <code>testCommentServer</code> and <code>authedCtx</code>